Contents

Chapter 1

Theoretical Background

This chapter provides an overview of the theoretical background underlying the main topics addressed in this thesis.
The first part of the chapter focuses on conventional step-index optical fiber, with the aim of explaining the physics of mode confinement. Starting from the wave equation, the electromagnetic field behavior in the fibers is derived. This leads, under the appropriate approximations, to the description of linearly polarized LP modes.
The second part of the chapter is devoted to the theory of optical dipole trapping. The physical mechanism that allows optical fields to confine neutral atom is discussed, and expressions for the trapping potential are derived for both ideal two-level atoms and realistic multi-level atoms.

1.1 Optical Fibers

Waveguides are devices that employ dielectric materials with a high refractive index to confine and guide electromagnetic radiation, preventing the diffraction-induced spreading that occurs during free-space propagation. The electromagnetic wave that propagates inside a waveguide is called guided mode.
Optical fibers are cylindrically symmetric waveguides, and represent the most widespread class of optical waveguides. Following chapter III of Yariv *et al.* [1], this section aims to explain the physical principles underlying light confinement in optical fibers. Particular attention is devoted to the characterization of the guided modes in the weakly guiding approximation, which leads to the so-called linearly polarized (LP) modes.

1.1.1 Circularly symmetric waveguides

An optical fiber can be modeled as a cylindrical core of refractive index n_1 surrounded by a cylindrical cladding of refractive index n_2. The refractive index distribution is therefore

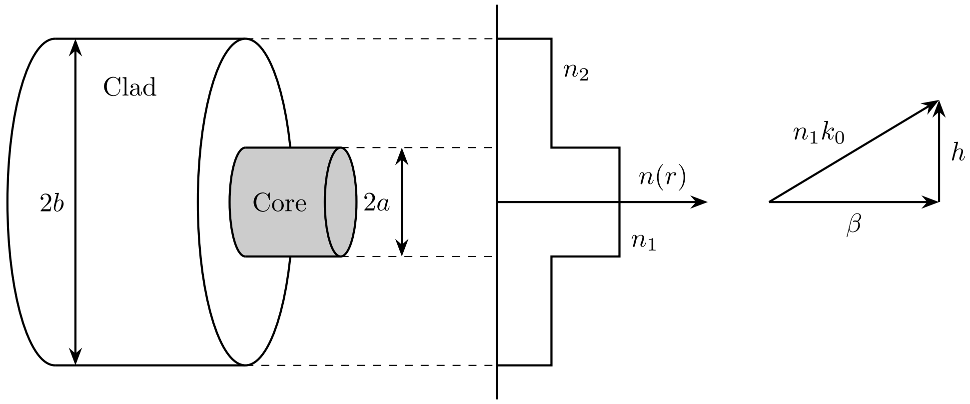

Figure 1.1: *Schematic representation of a step-index optical fiber (left), and wave vector diagram showing the geometric relationship between the wave number in the core n_1k_0, the longitudinal propagation constant β, and the transverse wave number h (right).*

described by the step-index profile

$$n(r) = \begin{cases} n_1, & 0 < r < a \\ n_2, & a < r < b \\ 1, & r > b, \end{cases} \tag{1.1}$$

where a is the radius of the core and b is the outer radius of the cladding. Usually the radius of the cladding is much larger than the radius of the core ($b \gg a$), thus the electromagnetic field can be considered zero at $r = b$. As an example, in a single-mode fiber guiding light with $\lambda = 1.5$ μm, the typical core radius is $a = 5$ μm while the cladding external radius is in the order of $b = 100$ μm.

1.1.1.1 The Cylindrical Wave Equation

Owing to the cylindrical symmetry of the system, it is convenient to work with the cylindrical coordinates:

$$\begin{cases} \mathbf{u_r} = & \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi \\ \mathbf{u}_\phi = & -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi \\ \mathbf{u_z} = & \hat{\mathbf{z}}, \end{cases} \tag{1.2}$$

thus the electric and magnetic fields are expressed in terms of the components E_r, E_ϕ, E_z, H_r, H_ϕ and H_z. In cylindrical coordinates the unit vectors $\mathbf{u_r}$ and $\mathbf{u}_\phi$ are not constant vectors ($\partial\mathbf{u_r}/\partial\phi = \mathbf{u}_\phi$ and $\partial\mathbf{u}_\phi/\partial\phi = -\mathbf{u_r}$). This complicates the wave equation for the associated components of the field. Nevertheless, as $\mathbf{u_z} = \hat{\mathbf{z}}$ the equation for the z components remains simple. Assuming the dependence on the coordinate z and time t to be $e^{i(\omega t-\beta z)}$, namely:

$$\begin{bmatrix}\mathbf{E}(\mathbf{r},t)\\ \mathbf{H}(\mathbf{r},t)\end{bmatrix} = \begin{bmatrix}\mathbf{E}(r,\phi)\\ \mathbf{H}(r,\phi)\end{bmatrix} e^{i(\omega t-\beta z)}, \tag{1.3}$$

where β is referred to as the propagation constant of the mode, the wave equation for E_z and H_z is

$$(\nabla^2 + k^2)\begin{bmatrix}E_z\\ H_z\end{bmatrix} = 0, \tag{1.4}$$

where $k^2 = \omega^2 n^2/c^2$ and ∇^2 is the Laplacian operator in cylindrical coordinates. To solve the cylindrical wave equation, the standard approach is to solve equation (1.4) and then express E_r, E_ϕ, H_r and H_ϕ in terms of E_z and H_z through Maxwell's curl equations in cylindrical coordinates (Appendix A of [1]).
Using (1.3) and explicitly writing the transverse cylindrical Laplacian in (1.4), one gets:

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} + (k^2-\beta^2)\right)\begin{bmatrix}E_z\\ H_z\end{bmatrix} = 0. \tag{1.5}$$

Equation (1.5) is solved by assuming the separation of variables

$$F_z(r,\phi) = R(r)\Phi(\phi), \tag{1.6}$$

where $F_z = E_z, H_z$, obtaining

$$\Phi\frac{d^2R}{dr^2} + \frac{1}{r}\Phi\frac{dR}{dr} + \frac{1}{r^2}R\frac{d^2\Phi}{d\phi^2} + (k^2-\beta^2)R(r)\Phi(\phi) = 0. \tag{1.7}$$

Dividing by $R\Phi$ and rearranging terms leads to

$$r^2\left(\frac{1}{R}\frac{d^2R}{dr^2} + \frac{1}{r}\frac{1}{R}\frac{dR}{dr} + (k^2-\beta^2)\right) = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}. \tag{1.8}$$

The last equation is in the form $f(r) = g(\phi)$. Therefore, since the left-hand side depends only on r and the right-hand side only on ϕ, both sides must be equal to the same constant:

$$\begin{cases} r^2\left(\frac{1}{R}\frac{d^2R}{dr^2} + \frac{1}{r}\frac{1}{R}\frac{dR}{dr} + (k^2-\beta^2)\right) = l^2 \\ -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = l^2 \end{cases}. \tag{1.9}$$

The angular part has a simple exponential solution:

$$\Phi(\phi) = C_1 e^{+il\phi} + C_2 e^{-il\phi}, \tag{1.10}$$

where C_1 and C_2 are constants that are determined by the boundary conditions of the problem. To be physically meaningful, $\Phi(\phi)$ must be single valued ($\Phi(\phi) = \Phi(\phi + 2\pi)$), thus the value of l is constrained to

$$l = 0, 1, 2, ... \tag{1.11}$$

A set of elementary solutions can therefore be expressed in the form

$$\begin{bmatrix} E_z \\ H_z \end{bmatrix} = R(r)e^{\pm il\phi},$$

where the differential equation that describes the radial part is

$$\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr} + \left(k^2 - \beta^2 - \frac{l^2}{r^2}\right) R = 0, \tag{1.12}$$

which is recognized as Bessel's differential equation, and its solutions are a linear combination of Bessel functions of order l.

1.1.1.2 The Bessel functions

The radial equation (1.12) admits different classes of solutions depending on the sign of the quantity $k^2 - \beta^2$:

- for $k^2 - \beta^2 > 0$, the solutions are given by the ordinary Bessel functions of the first and second kind of order l:

$$R(r) = \bar{C}_1 J_l(hr) + \bar{C}_2 Y_l(hr); \tag{1.13}$$

- for $k^2 - \beta^2 < 0$, the solutions are given by the modified Bessel functions of the first and second kind, of order l:

$$R(r) = \bar{C}_1 I_l(qr) + \bar{C}_2 K_l(qr), \tag{1.14}$$

where $h^2 = k^2 - \beta^2$ and $q^2 = \beta^2 - k^2$, and $\bar{C}_1$ and $\bar{C}_2$ are arbitrary constants determined by the boundary conditions.

The behavior of the ordinary and modified Bessel functions is illustrated in Figure 1.2. To proceed with the physical solution, it is useful to consider their asymptotic forms for both large and small arguments.

For $x \gg 1, l$, the asymptotic expressions are

$$\begin{aligned} J_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \cos\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right), \\ Y_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \sin\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right), \\ I_l(x) &\to \left(\frac{1}{2\pi x}\right)^{1/2} e^{x}, \\ K_l(x) &\to \left(\frac{\pi}{2x}\right)^{1/2} e^{-x}. \end{aligned} \tag{1.15}$$

For $x \ll 1$, the corresponding asymptotic expressions are

$$\begin{aligned} J_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^{l}, \\ Y_0(x) &\to \frac{2}{\pi}\left(\ln\frac{x}{2} + \gamma_E\right), \\ Y_l(x) &\to -\frac{(l-1)!}{\pi}\left(\frac{2}{x}\right)^{l} \quad l = 1, 2, 3, \ldots, \\ I_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^{l}, \\ K_0(x) &\to -\left(\ln\frac{x}{2} + \gamma_E\right), \\ K_l(x) &\to \frac{(l-1)!}{2}\left(\frac{2}{x}\right)^{l} \quad l = 1, 2, 3, \ldots, \end{aligned} \tag{1.16}$$

where γ_E is the Euler–Mascheroni constant.

These asymptotic behaviors play a crucial role in the construction of guided-mode solutions. In particular, the functions $Y_l(x)$ and $K_l(x)$ diverge as $x \to 0$, whereas $K_l(x)$ decays exponentially for large arguments. Consequently, only specific combinations of Bessel functions satisfy the physical requirements for confined modes.

Another important property is that the derivative of a Bessel function of order l can be expressed as a combination of Bessel functions of orders $l-1$ and $l+1$. Since the transverse field components E_r, E_ϕ, H_r, and H_ϕ are obtained from spatial derivatives of E_z and H_z, they are generally described by combinations of Bessel functions of different orders.

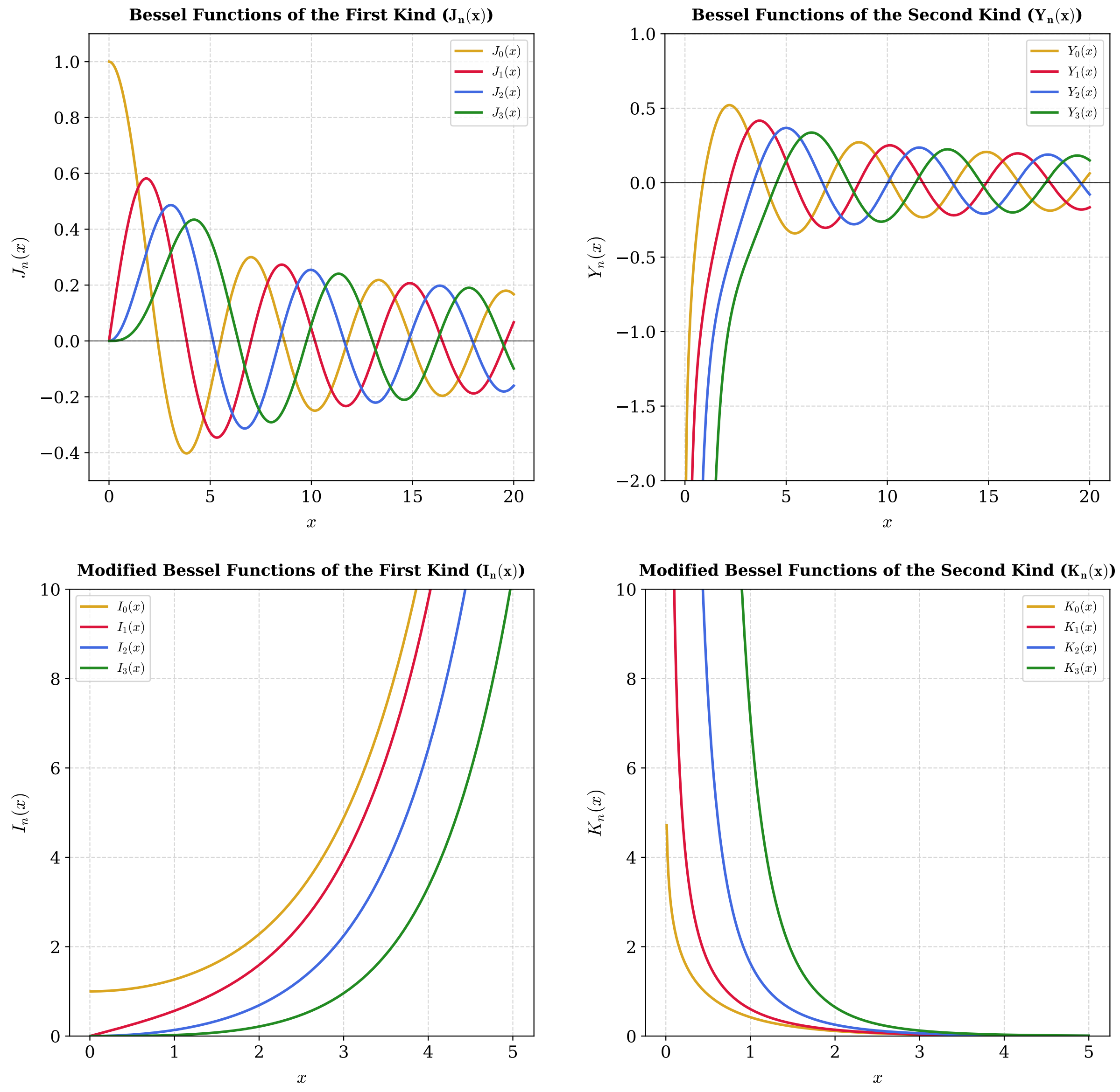

Figure 1.2: *Behavior of ordinary and modified Bessel functions for the first four integer orders, $n \in \{0, 1, 2, 3\}$. Top panels: Ordinary Bessel functions of the first kind, $J_n(x)$ (left), and second kind, $Y_n(x)$ (right). Bottom panels: Modified Bessel functions of the first kind, $I_n(x)$ (left), and second kind, $K_n(x)$ (right).*

1.1.2 The Weakly Guiding Approximation

The solution of the complete problem of electromagnetic guided modes in circular waveguides is cumbersome, and it leads to the so-called hybrid modes indicated by EH and HE, which are a combination of modes with the electric field transverse to the propagation of the wave (TE) and modes with the magnetic field transverse to the propagation direction (TM).

However, for a large variety of fibers, the core refractive index is only slightly higher than the refractive index of the cladding:

$$\frac{n_1 - n_2}{n_1} \ll 1, \tag{1.17}$$

a condition commonly referred to as weakly guiding approximation. Under this assumption, the exact modes can be accurately approximated by a simpler set of modes, known as linearly polarized (LP) modes, first introduced in the work of Gloge [2].

Using the relation $k^2 = (n(r)\,k_0)^2$ where $k_0 = \omega/c$, equation (1.5) can be rewritten as

$$[\nabla_t^2 + ((n(r)\,k_0)^2 - \beta^2)] \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.18}$$

Looking for a guided mode in the fiber is equivalent to seeking a solution that exhibits an oscillatory behavior in the core while exhibiting evanescent decay in the cladding. From the properties of second-order differential equations, these requirements imply:

$$\begin{aligned} r > a \quad &\text{vanishing} &&\Longrightarrow (n_2 k_0)^2 - \beta^2 < 0, \\ r < a \quad &\text{oscillating} &&\Longrightarrow (n_1 k_0)^2 - \beta^2 > 0. \end{aligned} \tag{1.19}$$

Consequently, the propagation constant of a guided mode must satisfy

$$n_2 k_0 < \beta < n_1 k_0, \tag{1.20}$$

which in weakly guiding regime becomes

$$n_2 k_0 \simeq \beta \simeq n_1 k_0. \tag{1.21}$$

At this point, it is convenient to introduce the transverse wave numbers in the core (h) and in the cladding (q):

$$h = \sqrt{n_1^2\,k_0^2 - \beta^2}, \tag{1.22}$$

$$q = \sqrt{\beta^2 - n_2^2\,k_0^2}. \tag{1.23}$$

The parameter h describes the transverse oscillation of the field within the core (see Figure 1.1), whereas q characterizes the decay of the mode in the cladding.
Equation (1.21) implies that

$$h, q \ll \beta. \tag{1.24}$$

Physically this means that the rays are approximately parallel to the fiber axis. Therefore, the longitudinal components of the electric and magnetic field are far weaker than the transverse ones, and the guided modes are approximately transverse electromagnetic (TEM) [3].

The objective is now to identify a complete set of approximate modal solutions in the weakly guiding limit. The approach is to focus on solutions where either the x or the y component of the field vanishes, thus looking for x-polarized or y-polarized solutions. The approximation made by Gloge [2] is based on postulating for the transverse component a simple dependence on a single Bessel function. Matching the requirements (1.19) with the Bessel equation solution condition (1.13) and (1.14), one deduces that the mode must be described by ordinary Bessel functions inside the core and by modified Bessel functions in the cladding. Furthermore, physically acceptable solutions must remain finite at the fiber axis and vanish at large radial distances. By comparison with the asymptotic behavior in (1.15) and (1.16), only J_l is admissible in the core region, whereas only K_l satisfies the required decay condition in the cladding.

For a mode polarized along the (y)-direction, the transverse electric field is therefore postulated as

$$\begin{aligned} E_x &= 0 \\ E_y &= \begin{cases} AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r < a \\ BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r > a \end{cases} , \end{aligned} \tag{1.25}$$

where A and B are constants that are determined from the continuity condition at the core–cladding interface.
It is worth noting that, for each value of l, equation (1.25) describes two independent azimuthal solutions associated to $e^{+il\phi}$ and $e^{-il\phi}$. Taking into account the two possible independent linear polarizations, a total of four independent basis functions are associated with each value of l.

The remaining electromagnetic field components can be obtained from Maxwell's equations. In particular, the magnetic field is given by [1]:

$$\begin{cases} H_x = \frac{-i}{\omega\mu}\frac{\partial}{\partial z}E_y = -\frac{\beta}{\omega\mu}E_y \\ H_y \simeq 0 \\ H_z = \frac{i}{\omega\mu}\frac{\partial}{\partial x}E_y \end{cases} , \tag{1.26}$$

and similarly, the longitudinal electric field component can be expressed as

$$E_z = \frac{i}{\omega\epsilon}\frac{\partial}{\partial y}H_x = \frac{-i\beta}{\omega^2\mu\epsilon}\frac{\partial}{\partial y}E_y. \tag{1.27}$$

Solving the differential equations for E_z and H_z yields the complete set of basis fields in the weakly guiding approximation [1].

In the core ($r < a$):

$$\begin{cases} E_x = 0 \\ E_y = AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \pm\frac{h}{\beta}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} + J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.28}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{ih}{\omega\mu}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} - J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.29}$$

In the cladding ($r > a$):

$$\begin{cases} E_x = 0 \\ E_y = BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \frac{q}{\beta}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} - K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.30}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{iq}{\omega\mu}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} + K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.31}$$

Considering that $E_z/E_y \propto h/\beta$, $H_z/H_x \propto h/\beta$ in the core and $E_z/E_y \propto q/\beta$, $H_z/H_x \propto q/\beta$ in the cladding, since $h, q \ll \beta$, E_y and H_x are the dominant components, and the mode is approximately TEM, consistently with the considerations given above.

Considering that $E_\phi = -E_x \sin(\phi) + E_y \cos(\phi)$, for a y-polarized mode $E_\phi \propto E_y$, thus the radial continuity condition at $r = a$ reduces to the continuity condition of E_y, which leads to

$$B = A\frac{J_l(ha)}{K_l(qa)}. \tag{1.32}$$

An identical condition is obtained from the continuity of H_ϕ.

On the other hand, the continuity condition of E_z at $r = a$ must hold for every value of the azimuthal coordinate ϕ. Consequently, the coefficients of $\exp[\phi(l+1)\phi]$ and $\exp[\pm i(l-1)\phi]$

must be matched separately. This leads to

$$ha\frac{J_{l+1}(ha)}{J_l(ha)} = qa\frac{K_{l+1}(qa)}{K_l(qa)}, \tag{1.33}$$

$$ha\frac{J_{l-1}(ha)}{J_l(ha)} = -qa\frac{K_{l-1}(qa)}{K_l(qa)}, \tag{1.34}$$

which are equivalent as a consequence of the recurrence relations satisfied by the Bessel functions [1]. The same characteristic equations are obtained by imposing continuity of H_z.

Finally, repeating the same procedure for an x-polarized mode leads to identical conditions, demonstrating that both polarizations share the same modal eigenvalues.

1.1.3 The LP modes

Equation (1.33) depends only on β through the transverse wave numbers h and q. It yields an infinite set of discrete solutions for each value of l. Therefore, the propagation constants are labeled as:

$$\beta_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.35}$$

Their corresponding modes are called Linearly Polarized modes, denoted by:

$$LP_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.36}$$

Since the condition in (1.33) holds for both x-polarized and y-polarized waves, the modes LP_{lm-x} and LP_{lm-y} are degenerate sharing the same propagation constant β_{lm}. This degeneracy is a direct consequence of the circular symmetry of the fiber. Furthermore, accounting for the degeneracy associated with the azimuthal phase term $e^{\pm il\phi}$, the propagation constant β_{lm} and the mode LP_{lm} are four-fold degenerate. A notable exception occurs for the fundamental case $l = 0$, for which the exponential is equal to unity independently of the sign of $l\phi$, reducing β_{0m} and LP_{0m} to a two-fold degeneracy.

1.1.3.1 Characteristic equation of the LP modes

In order to determine the allowed values of β_{lm}, the characteristic equation (1.33) must be solved. Since the transverse wave numbers h and q are interdependent, it is convenient to express them in terms of a dimensionless parameter. For this purpose, the normalized frequency, also called fiber V parameter, is introduced and defined as:

$$V = k_0 a\sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \sqrt{(ha)^2 + (qa)^2}. \tag{1.37}$$

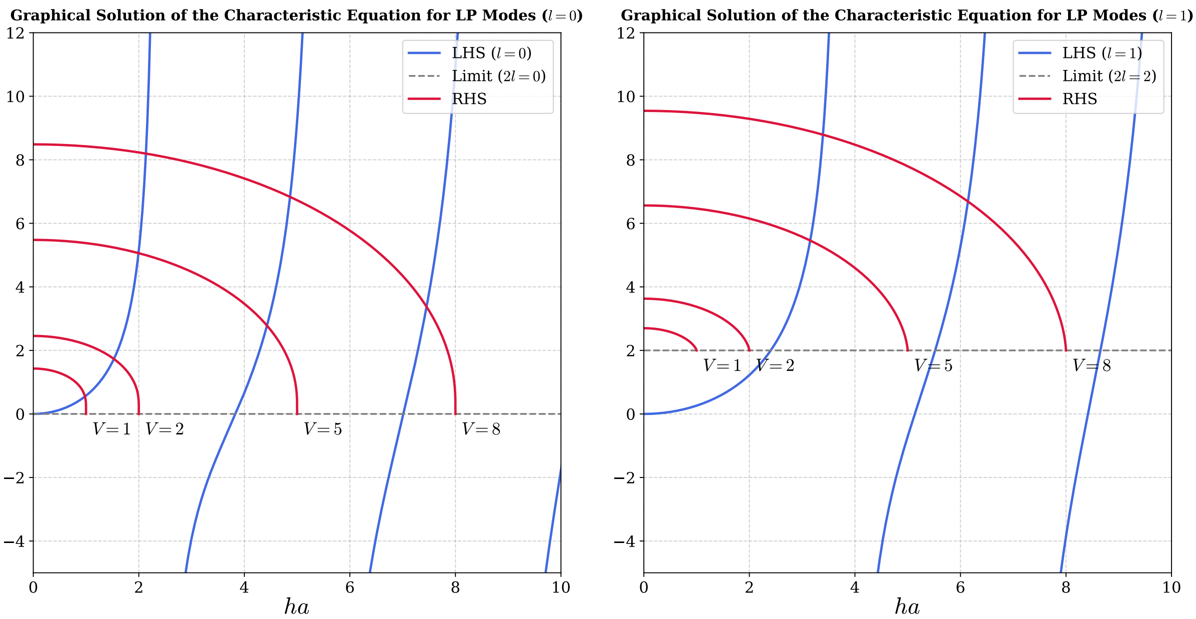

Figure 1.3: *Graphical solution of the characteristic equation* (1.33) *of the LP modes, for* $l = 0$ *(left) and* $l = 1$ *(right). The blue curves represent the left-hand side (LHS) of equation* (1.33) *for the given* l*. The red curves represent the right-hand side (RHS) for* $V = 2,\ 5,\ 8$*. The gray dashed line indicates the limit* $qa \to 0$ *of the LHS.*

The V parameter is an intrinsic property of the fiber determined by its structural and material characteristics. As shown below, it governs the number of guided modes and determines their respective confinement factor.

Using the V parameter, the cladding variable qa can be expressed as:

$$qa = \sqrt{V^2 - (ha)^2}, \tag{1.38}$$

and substituting it back in equation (1.33), the characteristic equation is reduced to a function of the single variable ha, allowing for a graphical or numerical solution. Moreover, equation (1.38) explicitly restricts the domain of the left-hand side of (1.33) to $ha < V$.

Figure 1.3 shows the graphical solution for $l = 0$ (left) and $l = 1$ (right). For $l = 0$, the left-hand side of equation (1.33) vanishes at the roots of $J_1(ha)$, located at $ha = 0,\ 3.832,\ 7.016\ \ldots$, whereas the denominator vanishes at the roots of $J_0(ha)$ which occur at $ha = 2.405,\ 5.520,\ 8.654\ \ldots$ corresponding to vertical asymptotes. Therefore, in the interval $ha \in [0, 2.405]$ the left-hand side is a strictly increasing positive function going from 0 to $+\infty$. Conversely, the right-hand side, at $ha = 0$ assumes the positive value $VK_1(V)/K_0(V)$ and monotonically decreases to 0 at $ha = V$. Consequently, an intersection point always exists, even for very small values of V, implying that the fundamental LP_{01} mode is always guided by an optical fiber regardless of its V parameter.

Using (1.16), one derives that for $l \geq 1$, the right-hand side asymptotically approaches $2l$ as $ha \to V$. Under these conditions, an intersection is no longer guaranteed for every

value of V. In particular for $V < 2.405$ the only possible guided mode is the LP_{01}. As V increases beyond 2.405, also the LP_{11} appears, therefore $V = 2.405$ is called the V cutoff value of the mode LP_{11}.
The cutoff value V_{cutoff} for a given mode can be determined by considering the condition at which the field is no longer evanescent in the cladding, corresponding to $q = 0$. Since $V = ha$ when $q = 0$, the cutoff condition can be obtained by imposing $q = 0$ in equation (1.34), yielding

$$V_{\text{cutoff}} \frac{J_{l-1}(V_{\text{cutoff}})}{J_l(V_{\text{cutoff}})} = 0. \tag{1.39}$$

For $l \geq 1$, in the limit $V_{\text{cutoff}} \to 0$ (see (1.16)) the left-hand side evaluates to $2l$ therefore for $l \geq 1$ the cutoff is strictly larger than zero. For $l = 0$ using the property $J_{-1}(V) = -J_1(V)$, for $V_{\text{cutoff}} \to 0$ one gets $0 = 0$, confirming that the mode LP_{01} is guided independently of the value of V.
Excluding the special case of the LP_{01}, equation (1.39) leads to

$$J_{l-1}(V_{\text{cutoff}}) = 0. \tag{1.40}$$

Therefore, the cutoff value of the LP_{lm} mode is given by the m-th zero of the Bessel function J_{l-1} if $l \neq 0$, and by the $(m-1)$-th zero of J_{l-1} if $l = 0$, while the LP_{01} mode is always guided. Here, the zero at the origin is not included in the enumeration of the zeros of the Bessel functions. The cutoff values of several low-order LP modes are reported in Table 1.1.
It is worth noting that equation (1.34) is employed to obtain the cutoff condition rather than (1.33) because its right-hand side conveniently vanished as $ha \to V$, making the cutoff directly associated to the roots of ordinary Bessel functions of the first kind.

Table 1.1: *Cutoff values of V for low-order LP modes.*

V_{cutoff}	$m = 1$	$m = 2$	$m = 3$	$m = 4$
$l = 0$	0	3.832	7.016	10.173
$l = 1$	2.405	5.520	8.654	11.792
$l = 2$	3.832	7.016	10.173	13.323
$l = 3$	5.136	8.417	11.620	14.796
$l = 4$	6.379	9.760	13.017	16.224

1.1.3.2 Power Flow of the LP modes

Prior to evaluating the power flow of the LP modes, a clarification regarding the basis chosen for the spatial representation of the field is necessary. So far, following the formalism in [1], a basis in which the azimuthal dependence is expressed by the exponential $e^{\pm il\phi}$, for the two degenerate solution $\pm l$ has been utilized. This representation is convenient for calculation, however, it leads to a basis in which the modes have an azimuthal phase dependence, or equivalently modes that carry a defined orbital angular momentum (OAM) $l\hbar$ per photon [4]. In typical experimental conditions, however, optical fibers are coupled with standard Gaussian beams, which carry zero macroscopic OAM due to their lack of azimuthal phase gradients. Therefore, to conserve the net angular momentum at the fiber input, the coupling excites the two azimuthal counter-propagating spatial modes $+l$ and $-l$ with identical amplitudes. Consequently, a basis employing trigonometric functions ($\cos(l\phi)$ and $\sin(l\phi)$) is physically more representative of typical intensity profiles and is widely adopted in the literature. These functions represent a linear superposition of counter-propagating azimuthal modes carrying opposite OAM, yielding a net-zero OAM state:

$$\begin{aligned} \cos(l\phi) &= \frac{e^{il\phi} + e^{-il\phi}}{2}, \\ \sin(l\phi) &= \frac{e^{il\phi} - e^{-il\phi}}{2i}. \end{aligned} \tag{1.41}$$

Therefore, the trigonometric basis is adopted for the remainder of this analysis. While a pure $e^{\pm il\phi}$ basis implies a ring-shaped intensity profile devoid of azimuthal modulation, the trigonometric basis captures the interference between opposite OAM states, producing the characteristic $2l$-lobe intensity patterns routinely observed experimentally.

The power flow of the LP modes is evaluated via the longitudinal component of the time-averaged Poynting vector:

$$S_z = \frac{1}{2}\,\mathrm{Re}[E_x H_y^* - E_y H_x^*]. \tag{1.42}$$

Using the field expressions (1.28), (1.29), (1.30) and (1.31), the trigonometric azimuthal basis, the power flux density becomes:

$$S_z^{(lm)} = \frac{\beta}{2\omega\mu} \begin{cases} |A|^2 J_l^2(hr)\cos^2(l\phi) & r < a \\ \\ |B|^2 K_l^2(qr)\cos^2(l\phi) & r > a \end{cases} , \tag{1.43}$$

where the dependence on m is implicitly contained in h and q. The power guided in the core and cladding is obtained by integrating the Poynting vector over the respective cross-sections (using infinite cladding approximation):

$$P_{\text{core}} = \int_0^{2\pi}\int_0^a S_z r\,dr\,d\phi, \tag{1.44}$$

$$P_{\text{clad}} = \int_0^{2\pi}\int_a^\infty S_z r\,dr\,d\phi. \tag{1.45}$$

Evaluating these integrals [1], and substituting B from (1.32), the result is:

$$P_{\text{core}} = \frac{\beta}{2\omega\mu}\pi a^2 |A|^2 \left[J_l^2(ha) - J_{l-1}(ha)J_{l+1}(ha)\right], \tag{1.46}$$

$$P_{\text{clad}} = \frac{\beta}{2\omega\mu}\pi a^2|A|^2 \left[-J_l^2(ha) - \left(\frac{h}{q}\right)^2 J_{l-1}(ha)J_{l+1}(ha)\right]. \tag{1.47}$$

Summing these contributions gives the total power flow:

$$P_{\text{tot}} = \frac{\beta}{2\omega\mu}\pi a^2|A|^2 \left(1 + \frac{h^2}{q^2}\right)[-J_{l-1}(ha)J_{l+1}(ha)]. \tag{1.48}$$

By multiplying equations (1.33) and (1.34) side by side, one gets:

$$(ha)^2\frac{J_{l+1}(ha)J_{l-1}(ha)}{J_l^2(ha)} = -(qa)^2\frac{K_{l+1}(qa)K_{l-1}(qa)}{K_l^2(qa)}. \tag{1.49}$$

Given that the modified Bessel functions K_l are always positive (see Figure 1.2), the quantity $J_{l-1}(ha)J_{l+1}(ha)$ is always negative for the allowed values of ha, therefore the total power flow (1.48) is always positive as physically expected.
The core confinement factor is defined as $\Gamma_1 = P_{\text{core}}/P_{\text{tot}}$

$$\Gamma_1 = \frac{P_{\text{core}}}{P_{\text{tot}}} = \frac{1}{V^2}\left((qa)^2 - \frac{(qa)^2J_l^2(ha)}{J_{l-1}(ha)J_{l+1}(ha)}\right). \tag{1.50}$$

Γ_1 gives a measure of how much of the mode is travelling in the core rather than in the cladding. This parameter is important for calculating the dispersion and attenuation properties of the fiber, which are critical parameters in the design of fibers for telecommunication systems. The dependence of Γ_1 on V is not trivial as both h and q depend on V through the solution of the characteristic equation (1.33). Generally, the confinement factor Γ_1 increases monotonically with V. The confinement factor at cutoff is obtained by taking the limit $q \to 0$ in (1.50), obtaining:

$$\Gamma_1 = \begin{cases} 0 & \text{for} \quad l = 0, 1 \\ 1 - \frac{1}{l} & \text{for} \quad l > 1 \end{cases}. \tag{1.51}$$

For example the mode LP_{21} at cutoff has a confinement efficiency of 0.5. Figure 1.4 illustrates the behavior of the confinement factor as a function of the V parameter for some low-order LP modes.

Finally, the calculated intensity profiles for selected LP modes are mapped in Figure 1.5. The figure highlights how the azimuthal index l governs the number of diametric nodal lines crossing the beam axis, whereas the radial index m dictates the number of concentric local intensity maxima along the radial coordinate.

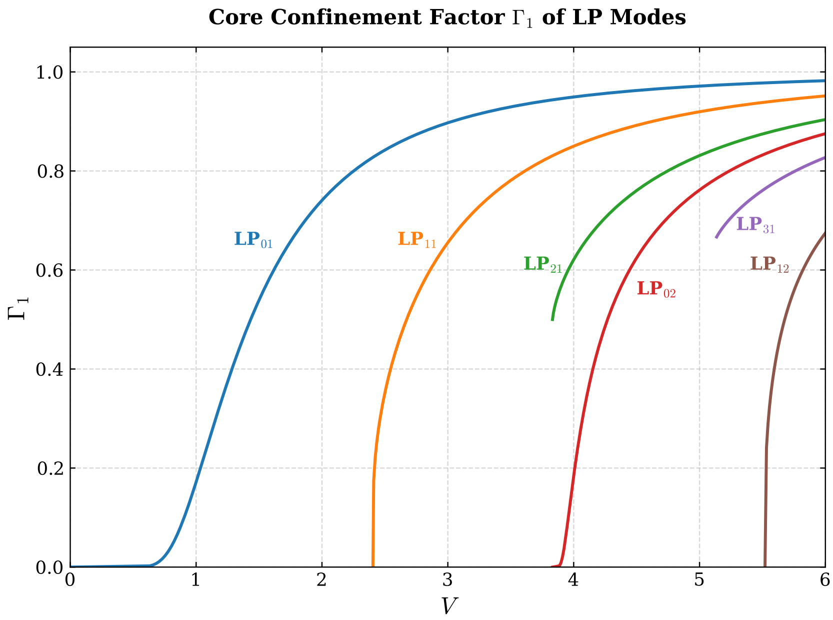

Figure 1.4: *Confinement factor* Γ_1 *of LP modes as a function of the fiber* V *parameter. Note that for the modes with* $l > 1$ *(*LP_{21}*,* LP_{31}*) the confinement at cutoff is different from* 0.

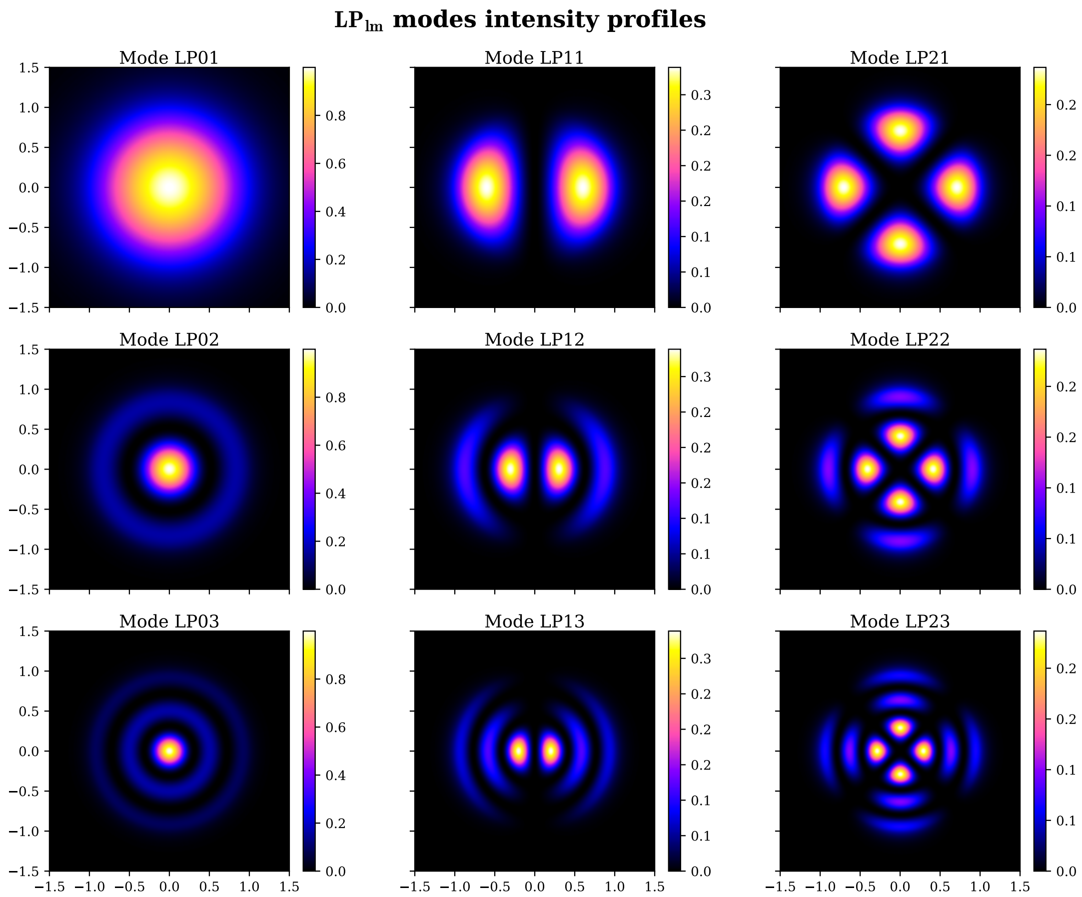

Figure 1.5: *Intensity profiles of the LP modes with* $l = 0, 1, 2$ *and* $m = 1, 2, 3$*. Each mode* LP_{lm} *is calculated with* $V^{(lm)} = V^{(lm)}_{cutoff} + 2$*. The* x, y *coordinates are expressed in units of fiber core radius.*

1.2 Hollow-Core Photonic Crystal Fibers

The previous section focussed only on light propagation in conventional step-index fibers. However, technological developments in the field of optical fibers have led to novel low-loss optical waveguides that are very promising for both telecommunication and fundamental research. Those are the so-called Photonic Crystal Fibers (PCFs) first introduced by Knight *et al.* [5]. For the purposes of this work, particular interest lies in the subclass of Hollow-Core Photonic Crystal Fibers (HCPCFs) that were introduced a few years later in 1999 by Cregan *et al.* [6].

In conventional fibers, light is confined within a solid core surrounded by a bulk cladding, requiring $n_{\text{core}} > n_{\text{clad}}$. By contrast, the core of an HCPCF is physically hollow (typically filled with air) and surrounded by a periodical microstructured cladding, resulting in $n_{\text{core}} < n_{\text{clad}}$. Figure 1.6 presents Scanning Electron Microscope (SEM) images illustrating various cladding microstructures utilized in HCPCFs.

This section proposes a review of the physical principles governing HCPCFs, based on the work of Debord *et al.* [7].

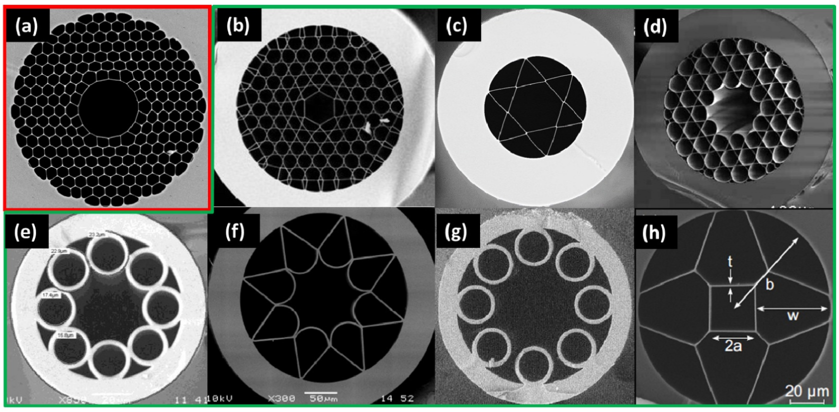

Figure 1.6: *SEM images of various hollow core fibers: (a) photonic band gap fiber; (b-h) inhibited coupling fibers. Credits [8].*

1.2.1 Introduction to HCPCF Guiding Principles

The guiding mechanism of conventional fibers can be explained employing the principle of total internal reflection (TIR), which strictly requires the condition $n_{\text{core}} > n_{\text{clad}}$. This condition is expressed in the wave optic treatment by Equation (1.20) as a requirement for a mode to be guided (oscillatory in the core and evanescent in the cladding). As previously mentioned, in an HCPCF $n_{\text{core}} < n_{\text{clad}}$, therefore the guiding mechanism must have a different physical nature.

The theoretical method used to explain the physics of these fibers is heavily drawn from condensed matter physics. In this approach, the periodic dielectric microstructure of the cladding (referred to as *photonic crystals*) is treated in analogy to a periodic potential in a crystal, while the core is treated as a defect of the periodicity. The propagation of guided light is mathematically solved by casting the electromagnetic wave equation as an eigenvalue problem in the frequency-wavevector space, in analogy to the solution in the k-space for electronic states in a crystal. Consequently, concepts and notations typical of quantum mechanics and condensed matter physics arise from the solutions, such as Bloch states, density of states and band gaps.

The crucial physical quantity for understanding the different guiding mechanisms is the modal spectrum of both the core and the cladding, represented by the Density Of Photonic States (DOPS) in the frequency-wavevector space (ω, β). Here, β is the propagation constant of the mode, namely the z component of the wave vector $\mathbf{k}$ of the mode, equivalent to the β defined in Section 1.1. In the literature, the DOPS is conventionally expressed in terms of the effective refractive index, $n_{\text{eff}} = \beta/k$.

Analogous to electronic band structures in solids, the DOPS (ω, n_{eff}) delineates regions where there are photonic states available corresponding to effective indices for which light can propagate, and void regions where there are no states available for light propagation. The latter regions are called Photonic Band Gaps (PBGs).

By comparing the core and cladding DOPS, two distinct guiding mechanisms emerge: The first is choosing a core index and a geometry so that at least a part of its supported mode (ω, n_{eff}) lies in the cladding PBG. In this way the core modes cannot leak out because there are no available modes in the cladding to excite at that specific effective index. The second relies on cladding modes and core modes that can coexist at the same n_{eff} but have a small spatial overlap so that the core modes cannot leak in the cladding because their coupling with the cladding mode continuum is suppressed.

These mechanisms divide the HCPCFs into two different classes: Photonic Band Gap (PBG) fibers and Inhibited Coupling (IC) fibers.

1.2.2 PBG and IC Guiding Mechanisms Comparison

Figure 1.7 summarizes the core and cladding modal content for the three different considered fiber architectures: TIR, PBG and IC. First, in TIR fibers the cladding is made of a uniform material with the refractive index of the glass n_{g}, therefore it can guide modes for $n_{\text{eff}} < n_{\text{g}}$, represented in Figure 1.7a as a continuum of cladding modes. Above this continuum no cladding modes exist effectively creating a gap. Since the core is made of a uniform material with the refractive index of doped glass $n_{\text{dg}} > n_{\text{g}}$, modes in the core exist for $n_{\text{eff}} < n_{\text{dg}}$, therefore, the guidance occurs for $n_{\text{g}} < n_{\text{eff}} < n_{\text{dg}}$. From this perspective, TIR fibers are just a form of PBG fibers, achieved with a core defect that has a uniform refractive index higher than that of the cladding.

Moving to a microstructured cladding, the requirement of a higher refractive index in the

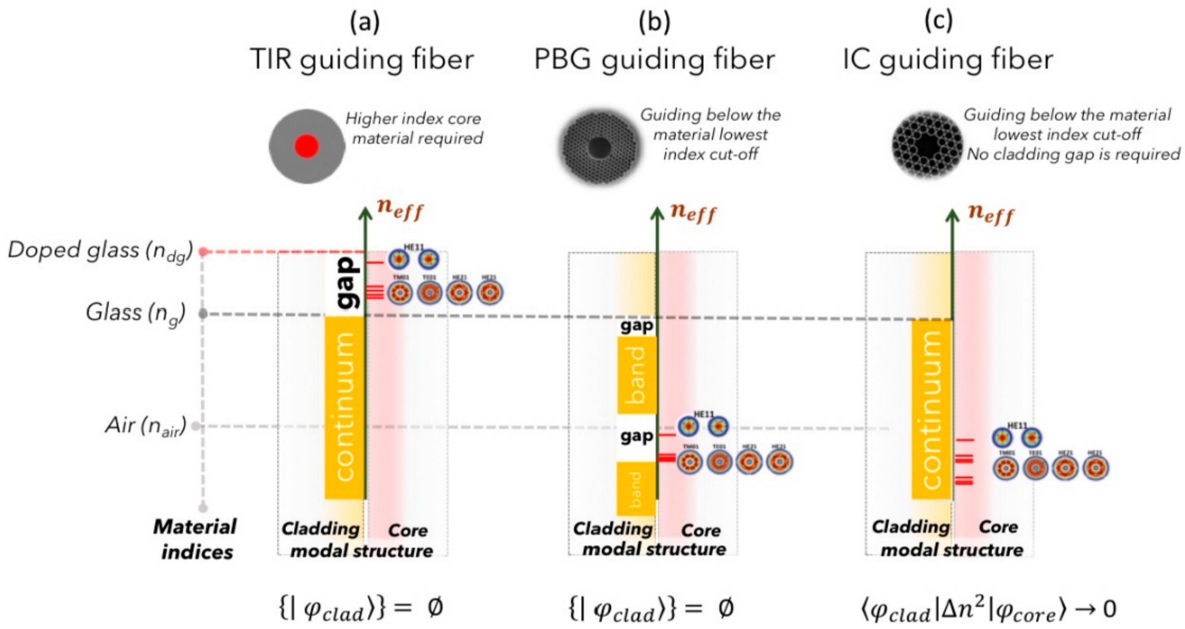

Figure 1.7: *Modal content representation of optical guiding fibers: (a) Total internal reflection (TIR); (b) Photonic Ban Gap (PBG); and (c) Inhibited Coupling (IC). The vertical axis represent the effective index n_{eff}, with the indices involved represented as horizontal dashed lines. Each graph represents the photonic bands of the cladding mode as a function of n_{eff}, with continuum (orange-filled rectangles) or gaps region (white-filled rectangles), highlighting where the core guidance happens. For both TIR and PBG fibers, guided core modes appear in correspondence of gaps. For IC fibers, no band gaps are present, and guidance occurs alongside the continuum of cladding modes. Credits [7].*

core is lifted. In this way the cladding photonic band structure acquires a more complex pattern made of an alternation of continuum regions and gap regions (Figure 1.7b). By a correct engineering of the microstructures, the gaps can range the region $n_{\text{eff}} < n_{\text{air}}$, allowing optical guidance in the air filled core.

Finally, IC fibers exhibit a fundamentally different guiding mechanism, as they are the only fiber type to not require a cladding PBG in correspondence of the core's guiding region (Figure 1.7c). In these fibers the cladding continuum and the core continuum of modes can coexist without leaking of the core modes to the cladding. The guiding mechanism is explained through the concept brought from quantum mechanics of Bound States in the Continuum [9, 10]. According to this model, the coupling between the field of the core mode, $|\phi_{\text{core}}\rangle$, and the cladding mode, $|\phi_{\text{clad}}\rangle$, is strongly reduced. Mathematically, this translates to the condition $\langle\phi_{\text{clad}}|\Delta n|\phi_{\text{core}}\rangle \longrightarrow 0$, with Δn being the transverse index profile function [7]. This decoupling can be achieved by either having little spatial intersection between the fields of the $|\phi_{\text{clad}}\rangle$ and $|\phi_{\text{core}}\rangle$ photonic states, or by having a strong mismatch in their respective transverse spatial phases. This low overlap also explains why IC-HCPCF, among the fibers, are those that achieve the lowest loss. Notably, since the guidance occurs at a continuum of cladding modes, the IC fibers do not have

any cutoff frequency, therefore, their nature is intrinsically multimodal.
To further reduce the overlap between core and cladding modes, Wang *et al.* [11] introduced an optimized core-contour shape known as a hypocycloid. Also referred to as a negative-curvature core contour, it consists of a series of alternating negative-curvature cusps defined by an inner radius R_{in} and an outer radius R_{out}. Figure 1.8 shows the difference between a standard core kagome IC-HCPCF and a hypocycloid variant. In particular Figure 1.8 right shows a hypocycloid kagome 7-cell (K7) fiber, very similar to the one employed in this work (see Section 2.2.3).

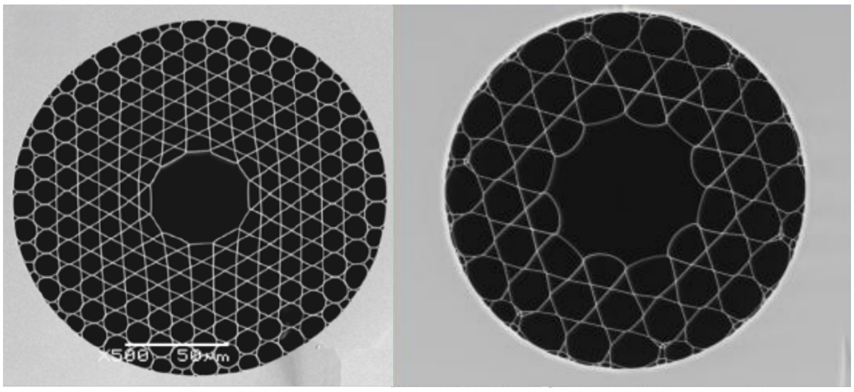

Figure 1.8: *Comparison between the SEM images of a kagome IC-HCPCF with standard core-contour (left) and a kagome IC-HPCF with hypocycloid core-contour (right). Credits [12, 7]*

1.2.3 HCPCF core guided modes

Despite the substantial physical difference in the guiding mechanisms of PBG and IC with respect to the TIR, the core modes do not deviate significantly from the modes in conventional step-index optical fibers [7]. More precisely, the propagation constants and the mode fields do not differ much from the results obtained by Marcatili and Schmeltzer [14] for electromagnetic modes in hollow dielectric capillaries, which, in turn, share the same qualitative shapes as the complete solution TE, TM, EH and HE modes of step-index optical fibers. For instance, Digonnet *et al.* [15] show how the $LP_{l,m}$ modes remain a good approximation for the core mode of PBG fibers and LP mode based approximation can effectively yield estimates for the number of supported modes, their cutoffs, and their effective indices. However, for more accurate, quantitative predictions and a rigorous description, the complete solutions must be computed [16].
Figure 1.9 left presents the calculated LP-like modes for a kagome IC-HCPCF from [7]. The modal composition of a fiber can be experimentally characterized using Spatially and Spectrally resolved imaging (S^2) technique, which was first introduced by Nicholson *et al.* [17]. This technique relies on the acquisition and analysis of the optical interference spectrum generated by the superposition of multiple guided modes. By applying a Fourier

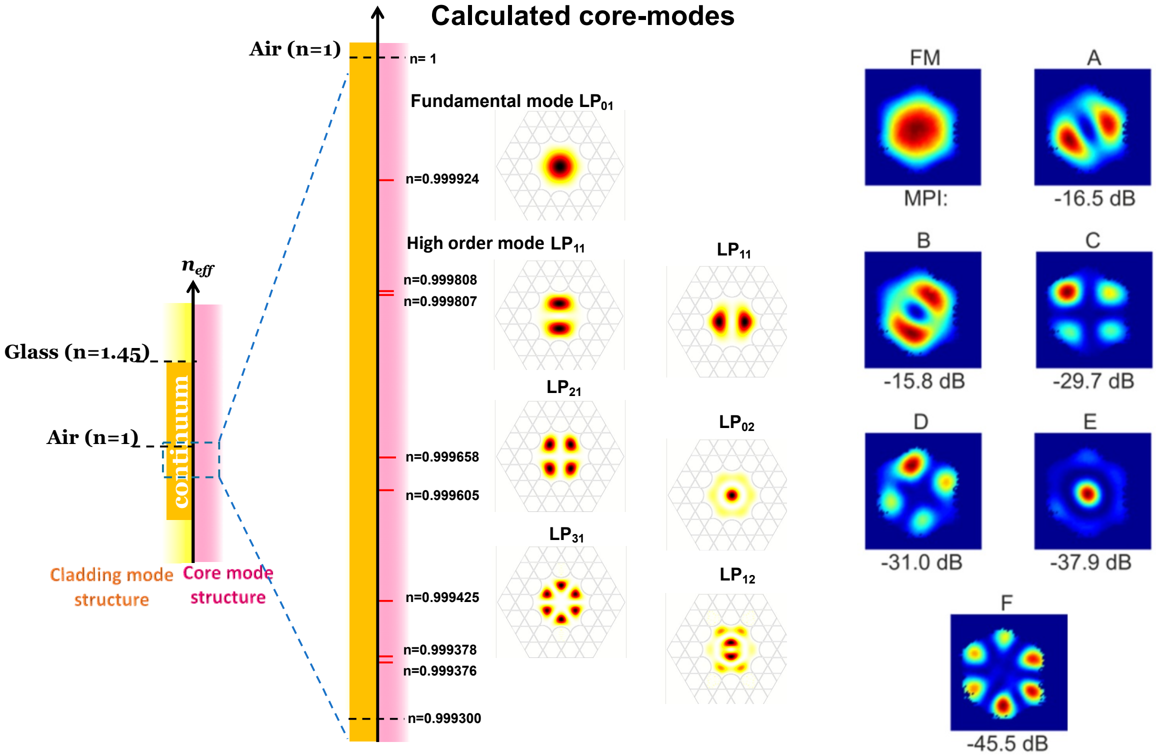

Figure 1.9: *LP-like modes in HCPCFs. Left: effective refractive index spectrum and intensity profiles of the calculated LP modes for the core of a Kagome fiber. Credits [7]. Right: modal profiles experimentally measured via the S^2 technique on a 7-cell (K7) Kagome fiber, with the corresponding Multipath Interference (MPI). The image reports the fundamental mode (FM); two different orientation of LP_{11}-like modes (A, B); two different orientation of LP_{12}-like modes (C,D); an LP_{02}-like mode (E); and an LP_{31}-like mode (F). Credits [13].*

transform to the measured spectrum, the differential group delays between the propagating modes can be extracted, enabling the simultaneous characterization of both their spatial profiles and their spectral properties. Figure 1.9 right displays the experimentally measured mode profiles for a K7 IC-HCPCF, obtained using the S^2 technique by Bradley *et al.* [13]. The measured mode profiles are clearly similar to both the LP modes of the conventional step-index fibers, and to the theoretically calculated ones in Figure 1.9 left. Beneath each mode profile, Figure 1.9 right also reports the value of the Multipath Interference (MPI), defined as the ratio, expressed in decibels, of the power guided by the high-order mode under consideration (P_{HOM}) to the power guided by the fundamental mode (P_{FM}): $\mathrm{MPI} = 10\log(P_{\mathrm{HOM}}/P_{\mathrm{FM}})$. It is worth noting that the guided light is mainly in the fundamental core mode as the lowest MPI, represented by the LP_{11}-like modes, which, when combined, yield ~ -13 dB, indicating that the high-order modes carry roughly 1/20 of the total transmitted power. According to [7], an IC-HCPCF can effectively operate as a single-mode fiber if engineered to exhibit high losses for higher-order modes while maintaining low loss for the fundamental mode.

1.3 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm *et al.* [18], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.3.1 Oscillator model

The optical dipole force arises from the interaction of the induced electric dipole moment with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.3.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \ , \tag{1.52}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.53}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.54}$$

Averaging over a field period [18]:

$$U_{\text{dip}} = -\frac{1}{2}\langle \mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha]I, \tag{1.55}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.

It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}} = \frac{1}{2\epsilon_0 c}\text{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.56}$$

The dipole oscillation absorbs a power

$$P_{\text{abs}} = \langle \dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega\text{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c}\text{Im}[\alpha]I, \tag{1.57}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{P_{\text{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\text{Im}[\alpha]I(\mathbf{r}). \tag{1.58}$$

1.3.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.59}$$

Using $E(t) = \mathcal{E}_0\exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0\exp(-i\omega t)$, equation (1.59) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.60}$$

Using $p = -ex(t) = -ex_0\exp(-i\omega t)$ and (1.60), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.61}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.62}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\, \Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.63}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3} |\langle e|\, \hat{d}\, |g\rangle|^2, \tag{1.64}$$

where $\hat{d}$ is the dipole moment operator.
Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.59 (for a rigorous derivation see Chapter V of Steck [19]). Therefore, the classical polarizability given by equation 1.63 represent a well suited approximation for ODTs.

1.3.1.3 Dipole potential and scattering rate

Substituting α from (1.63) in (1.55) and (1.58) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit functions of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right) I(\mathbf{r}), \tag{1.65}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3 \left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.66}$$

These equations show two resonance frequencies for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, in typical experimental conditions, it holds that $|\Delta| \ll \omega_0$. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta} I(\mathbf{r}) \tag{1.67}$$

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\mathrm{dip}} \tag{1.68}$$

Equations (1.67) and (1.68) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/Δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.3.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.3.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the electric dipole interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}}\cdot\mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j\neq i}\frac{|\langle j|\mathcal{H}_I|i\rangle|}{\varepsilon_i - \varepsilon_j}. \tag{1.69}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + (n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.69) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.70}$$

where Δ_{ij} is the detuning of the $i \to j$ transition with respect to the laser field.
Considering a two-level atom first, (1.69) has only one term, using equations (1.64) and

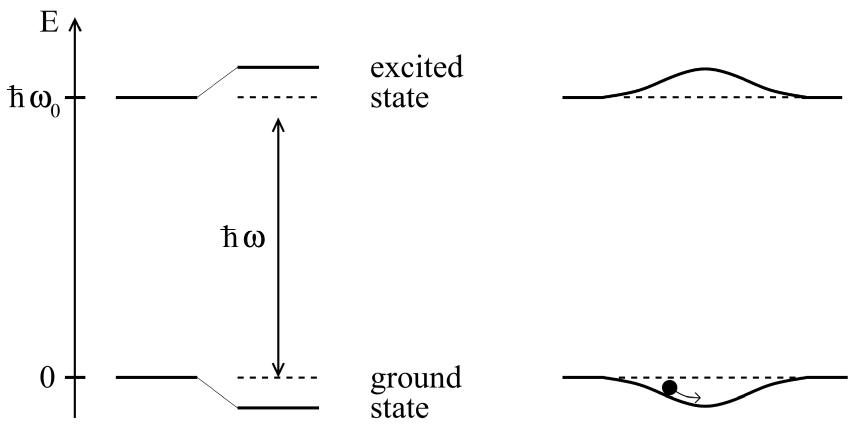

Figure 1.10: *Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms. Credits [18].*

(1.70) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm\frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm\frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.71}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.71) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.67). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. Assuming a low saturation regime, corresponding to $s \ll 1$, where the saturation parameter s is given by

$$s = \frac{\Omega^2/2}{\Delta^2 + (\Gamma^2/4)}, \tag{1.72}$$

and Ω is the Rabi frequency, defined as

$$\Omega = \frac{|\mathcal{E}_0|}{\hbar}\langle e|\,\hat{d}\,|g\rangle\,, \tag{1.73}$$

the atomic population can be considered almost entirely confined to the ground state. Consequently, the light-shifted ground state becomes the effective potential that governs the motion of the atoms, driving them toward regions of minima of ΔE (Figure 1.10).

For a multi-level atom Equation (1.69) contains many terms similar to equation (1.71), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. Each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\,\hat{d}\,|e_j\rangle$.

Applying the Wigner–Eckart theorem [20], this matrix element can be factored as:

$$d_{ij} = c_{ij}||d|| \tag{1.74}$$

where $||d||$ is the reduced matrix element given by (1.64) and c_{ij} are real Clebsch-Gordan coefficients describing the selection rules and coupling strengths of the transitions. They depend on the light polarization and the angular momentum quantum numbers of the involved hyperfine states.
Therefore, the energy shift (1.69) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.75}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.76}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transitions, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.3.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the visible and infrared (VIS-IR) range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structures are particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.
When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is $^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

- D_1: $^2S_{1/2} \to {}^2P_{1/2}$;
- D_2: $^2S_{1/2} \to {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting

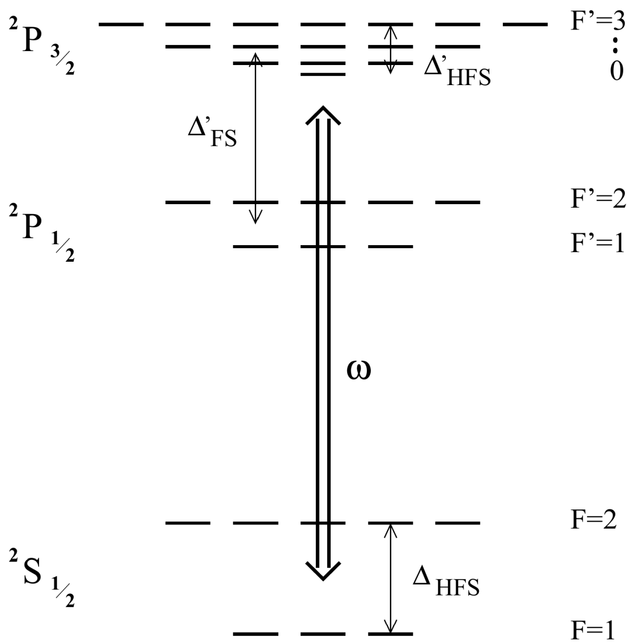

Figure 1.11: *Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb. Credits [18].*

as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.11.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.75) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [18]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P} g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P} g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.77}$$

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^{\pm}$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg \Delta'_{FS}$), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.77) at first order in the small parameter Δ'_{FS}/Δ, one gets [18]:

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3} \frac{\Gamma}{\Delta} \left(1 + \frac{1}{3} \mathcal{P} g_F m_F \frac{\Delta'_{\text{FS}}}{\Delta} \right) I(\mathbf{r}). \tag{1.78}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.67)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.78) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [18]. Therefore, linearly polarized light is usually preferred for dipole trap implementation.

1.3.3 Heating mechanisms

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.3.3.1 Heating source

In general, when atoms interact with light and absorb photos, when the subsequent spontaneous emission event occur, the randomness of the emitted photons' direction causes a diffusion in the phase space that corresponds to a heating processes. This discrete momentum transfer is the fundamental phenomenon that establishes lower limits for laser cooling mechanisms [21, 22].

At large detuning the scattering process follows a Poisson statistic. In this regime, the absorption of a photon from the dipole beam introduces a heating energy equivalent to the recoil energy $E_{\text{rec}} = k_B T_{\text{rec}}/2$ per scattering event, where T_{rec} is the recoil temperature. Since the absorptions occur strictly along the propagation axis the momentum diffusion occurs only along this axis, making the associated heating effect anisotropic. On the other hand, the subsequent spontaneous emission also increase the thermal energy by one recoil energy $E_{\rm rec}$ per event, however the emission directions are random, thus the effect is spread over all the three spatial direction.

Therefore, in a single scattering cycle, the atom gains $E_{\rm rec}$ of thermal energy from the absorption and then $E_{\rm rec}$ from the spontaneous emission, for a total of $2E_{\rm rec}$, in a time $\bar{\Gamma}_{\rm sc}^{-1}$. It leads to the total heating power

$$P_{\rm heat} = 2E_{\rm rec}\bar{\Gamma}_{\rm sc} = k_B T_{\rm rec}\bar{\Gamma}_{\rm sc}, \tag{1.79}$$

the heating depends on the mean scattering rate $\bar{\Gamma}_{\rm sc}$ and on the recoil $E_{\rm rec}$ which depends on the laser frequency.

Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [18].

In addition to the fundamental heating mechanism, it must be mentioned also the presence of technical heating caused by fluctuations in the laser power and in its pointing direction. In particular power fluctuations at twice the characteristic trap frequencies are relevant, as they can parametrically drive the oscillatory atomic motion [23].

1.3.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy, and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.80}$$

Using (1.79) in (1.80), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3}T_{\rm rec}\bar{\Gamma}_{\rm sc}. \tag{1.81}$$

Expressing the dipole potential as

$$\bar{U}_{\rm dip} = U_0 + \bar{E}_{\rm pot} = U_0 + \frac{3}{2}k_B T, \tag{1.82}$$

where U_0 is the potential energy minimum.

Using equation (1.68), $\bar{U}_{\rm dip}$ can be expressed as:

$$\bar{U}_{\rm dip} = \frac{\hbar\Delta\Gamma_{\rm sc}}{\Gamma}. \tag{1.83}$$

Substituting this into Equation (1.82) yields the mean scattering rate:

$$\bar{\Gamma}_{\mathrm{sc}} = \frac{\Gamma}{\hbar\Delta}(U_0 + \frac{3}{2}k_B T). \tag{1.84}$$

It is important to notice that, for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, by contrast for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.84) in (1.81), with a depth $\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\mathrm{red}} = \frac{1}{3}T_{\mathrm{rec}}\frac{\Gamma}{\hbar|\Delta|}\hat{U}, \tag{1.85a}$$

$$\dot{T}_{\mathrm{blue}} = \frac{1}{2}T_{\mathrm{rec}}\frac{\Gamma}{\hbar\Delta}k_B T. \tag{1.85b}$$

Equations (1.85) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

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